Financial Derivatives
ES32079

Lecture 2

Types of Traders and Hedging

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Structure of this lecture

Today we cover the following topics:

1. Types of Traders
2. Hedging
3. Hedging Equity Portfolios

Reference: Hull chapters 1 and 3.

Why Do Traders Use Derivatives?

Three important motives

- **Hedging:** reducing an existing exposure to risk
- **Speculation:** taking risk in anticipation of a price movement
- **Arbitrage:** exploiting inconsistent prices in different markets

Important distinction

These categories describe a trader's **objective**, not the instrument being traded. The same derivative may be used for hedging, speculation, or arbitrage.

Speculating on a Share-Price Increase

An investor has \$2,000 and expects a share price to rise over the next two months.

Current share price	\$20.00
Two-month call strike, K	\$22.50
Call premium per share	\$1.00

Two possible strategies

1. Buy \$2,000/\$20 = 100 shares.
2. Buy 20 call contracts, each covering 100 shares. This gives exposure to 2,000 shares and costs 2,000(\$1) = \$2,000.

Shares or Calls: Which Strategy Performs Better?

Let S_T be the share price after two months.

Profit from 100 shares

$$\Pi_T^{\text{shares}} = 100(S_T - 20).$$

Profit from calls covering 2,000 shares

$$\Pi_T^{\text{calls}} = 2{,}000\,[\max(S_T - 22.50, 0) - 1]\,.$$

Calculate

Find the profit from each strategy when $S_T = \$15$, $\$22.50$, and $\$25$.

Speculative Strategies: Results

S_T	Share profit	Call profit
\$15.00	−\$500	−\$2,000
\$22.50	\$250	−\$2,000
\$25.00	\$500	\$3,000

Interpretation

Options provide leverage: they can generate a larger percentage gain, but the buyer can lose the entire premium. The call strategy breaks even when

$$S_T = K + c_0 = \$22.50 + \$1.00 = \$23.50.$$

Can the Same Share Have Two Different Prices?

Suppose the following prices are observed simultaneously:

- London share price: $\$100$
- New York share price: \$150
- Exchange rate: 1.5300 USD per GBP

Compare prices in the same currency

$$\$100 \times 1.5300 = \$153.$$

The share is \$3 cheaper in New York.

Arbitrage strategy

Buy the share in New York for \$150 and simultaneously sell it in London for the equivalent of \$153:

$$\text{profit per share} = \$153 - \$150 = \$3.$$

Ignoring transaction costs, taxes, bid–ask spreads, trading restrictions, and settlement

Protecting a Share Portfolio with Put Options

An investor owns 1,000 shares currently worth \$28 each. A two-month put with strike $K = \$27.50$ costs \$1 per share.

The hedge

The investor buys 10 put contracts, each covering 100 shares:

$$10 \times 100 = 1{,}000 \text{ protected shares.}$$

The total premium is

$$1{,}000(\$1) = \$1{,}000.$$

Question

What happens to the combined position if the share price falls to \$20?

How Much Protection Does the Put Provide?

At maturity, the value of the hedged position is

$$V_T = 1{,}000 S_T + 1{,}000 \max(27.50 - S_T, 0).$$

S_T	Share value	Put payoff	Hedged value
\$20.00	\$20,000	\$7,500	\$27,500
\$27.50	\$27,500	\$0	\$27,500
\$35.00	\$35,000	\$0	\$35,000

Interpretation

The put establishes a minimum terminal value of \$27,500. Protection is not free: the investor paid a \$1,000 premium initially.

Long and Short Futures Hedges

Long hedge

Use a long futures position when an asset will be **purchased** later and the hedger is concerned that its price may rise.

Short hedge

Use a short futures position when an asset will be **sold** later and the hedger is concerned that its price may fall.

Classify each exposure

1. An airline will buy fuel in three months.
2. A farmer will sell wheat after the harvest.
3. A UK importer must buy US dollars in two months.
4. A fund plans to sell part of its equity portfolio.

Short Hedge for the Sale of an Asset

A hedger will sell an asset at t_2 and shorts futures at t_1. Let F_1 be the initial futures price, and let S_2 and F_2 be the spot and futures prices when the hedge is closed.

Effective selling price

The asset is sold for S_2, and the short futures position earns $F_1 - F_2$. Therefore,

$$S_2 + (F_1 - F_2) = F_1 + b_2, \qquad b_2 = S_2 - F_2.$$

Interpretation

If b_2 were known when the hedge began, the effective selling price would be known. Uncertainty about b_2 creates basis risk.

Long Hedge for the Purchase of an Asset

A hedger will buy an asset at t_2 and buys futures at t_1.

Effective purchase price

The asset costs S_2, and the long futures position earns $F_2 - F_1$.
Subtracting the futures gain from the cash purchase price gives

$$S_2 - (F_2 - F_1) = F_1 + b_2, \qquad b_2 = S_2 - F_2.$$

Key point

Long and short hedges produce the same algebraic effective price, $F_1 + b_2$, but they protect opposite commercial exposures.

A Numerical Example of Basis Risk

A producer shorts futures at $F_1 = \$52$. When the hedge is closed,

$$S_2 = \$47, \qquad F_2 = \$48.$$

Effective selling price

The cash-market sale produces \$47 and the futures gain is

$$F_1 - F_2 = 52 - 48 = \$4.$$

Hence,

$$\text{effective selling price} = 47 + 4 = \$51.$$

Equivalently,

$$F_1 + b_2 = 52 + (47 - 48) = \$51.$$

Understanding Basis Risk

The basis is

$$b_t = S_t - F_t.$$

- The basis may strengthen or weaken while the hedge is open.
- The closing basis determines the effective hedged price.
- For the same underlying and delivery terms, spot and futures prices converge at delivery, so $b_T = 0$.
- Cross hedges generally retain basis risk because the exposure and futures contract reference different assets.

Choosing a Futures Contract

1. Choose an underlying whose price is closely related to the exposure.
2. Choose a delivery month close to, but normally later than, the end of the hedge.
3. Check the contract size and calculate the required number of contracts.
4. Consider liquidity, bid–ask spreads, and basis risk.

Cross hedge

When no futures contract exists on the asset being hedged, use a futures contract on a related asset whose price changes are strongly correlated with the exposure.

Deriving the Minimum-Variance Hedge Ratio

For a hedged exposure, write the change in value per unit as

$$\Delta V_H = \Delta S - h\Delta F.$$

Its variance is

$$\text{Var}(\Delta V_H) = \sigma_S^2 + h^2\sigma_F^2 - 2h\rho\sigma_S\sigma_F.$$

Minimising with respect to h,

$$\frac{\partial \text{Var}(\Delta V_H)}{\partial h} = 2h\sigma_F^2 - 2\rho\sigma_S\sigma_F = 0.$$

Therefore,

$$\boxed{h^* = \rho\frac{\sigma_S}{\sigma_F}}.$$

Interpreting the Optimal Hedge Ratio

In

$$h^* = \rho \frac{\sigma_S}{\sigma_F},$$

σ_S is the standard deviation of spot-price changes, σ_F is the standard deviation of futures-price changes, and ρ is their correlation.

- A higher correlation generally makes the hedge more effective.
- h^* need not equal one.
- The hedge minimises variance; it does not necessarily eliminate risk.

Regression interpretation

In the regression

$$\Delta S_t = \alpha + h \Delta F_t + \varepsilon_t,$$

the estimated slope is the minimum-variance hedge ratio.

Optimal Number of Futures Contracts

If Q_A is the size of the exposure and Q_F is the quantity covered by one futures contract, then

$$\boxed{N^* = h^* \frac{Q_A}{Q_F}}.$$

The futures position covers the equivalent of

$$N^* Q_F = h^* Q_A$$

units of the exposure.

Important

This is a minimum-variance hedge, not necessarily a perfect hedge. Contract rounding and basis risk leave some residual exposure.

Cross-Hedging an Airline's Fuel Purchase

An airline expects to purchase 2 million gallons of jet fuel in one month. There is no suitable jet-fuel futures contract, so the airline considers heating-oil futures.

- Exposure: $Q_A = 2{,}000{,}000$ gallons
- Heating-oil contract size: $Q_F = 42{,}000$ gallons
- $\sigma_S = 0.0263$
- $\sigma_F = 0.0313$
- $\rho = 0.928$

Before calculating

Should the airline take a long or short futures position?

Airline Hedge: Calculation

The minimum-variance hedge ratio is

$$h^* = 0.928 \left(\frac{0.0263}{0.0313} \right) \approx 0.780.$$

The optimal number of contracts is

$$N^* = 0.780 \left(\frac{2{,}000{,}000}{42{,}000} \right) \approx 37.14.$$

Decision

The airline should take a **long position in approximately 37 heating-oil futures contracts**. A long hedge protects against an increase in the fuel price.

Residual risk

Why is the hedge imperfect even before the number of contracts is rounded?

Stock Indices and Index Futures

A stock index measures the performance of a specified portfolio of shares. Different indices use different weighting rules:

- S&P 500: float-adjusted market-capitalisation weighted
- Nasdaq-100: modified market-capitalisation weighted
- Dow Jones Industrial Average: price weighted

CME S&P 500 futures examples

- E-mini S&P 500: \$50$\times$ the index
- Micro E-mini S&P 500: \$5$\times$ the index

For an index level of 5,000, the respective contract notionals are \$250,000 and \$25,000.

Hedging a Portfolio That Mirrors an Index

If a portfolio moves one-for-one with the index, the number of futures contracts needed for a full hedge is

$$N^* = \frac{V_A}{V_F},$$

where V_A is the portfolio value and V_F is the notional value of one futures contract.

Example

A \$5 million portfolio mirrors the S&P 500. If an E-mini contract has a notional value of \$250,000,

$$N^* = \frac{5{,}000{,}000}{250{,}000} = 20.$$

To protect the long portfolio against a market fall, short 20 contracts.

Market Risk and Beta

A portfolio does not normally mirror an index exactly. Its sensitivity to market movements can be represented by beta:

$$r_P - r_f = \alpha + \beta_P(r_M - r_f) + \varepsilon_P.$$

- β_P: exposure to systematic market risk
- ε_P: idiosyncratic return not explained by the market
- $\beta_P = 1$: portfolio tends to move with the market
- $\beta_P > 1$: greater sensitivity to market movements

Limitation

Beta is estimated from historical data and may change. Index futures hedge systematic risk, not the portfolio's idiosyncratic risk.

Changing a Portfolio's Beta with Futures

To change a portfolio beta from β_P to a target β^*, use

$$\boxed{N^* = (\beta_P - \beta^*)\frac{V_A}{V_F}}.$$

- If $\beta_P > \beta^*$, short futures.
- If $\beta_P < \beta^*$, buy futures.
- Setting $\beta^* = 0$ removes the estimated market exposure.

Special case: beta-neutral hedge

$$N^* = \beta_P \frac{V_A}{V_F}.$$

A Numerical Beta Hedge

A portfolio is worth \$5 million and has $\beta_P = 1.20$. The E-mini S&P 500 futures price is 5,000, so

$$V_F = \$50(5{,}000) = \$250{,}000.$$

To reduce the portfolio beta to zero,

$$N^* = (1.20 - 0)\left(\frac{5{,}000{,}000}{250{,}000}\right) = 24.$$

Decision

Short 24 E-mini S&P 500 futures contracts.

Extension

How many contracts would be required to reduce the beta to 0.50?

Why Use a Beta Hedge?

An investor may have a view about particular shares but not about the overall market.

Example

An investor expects a portfolio of biopharmaceutical shares to outperform the market but is concerned that the entire market may fall. Short index futures can reduce the broad market exposure while retaining the portfolio's relative performance exposure.

What remains after the hedge?

The hedge reduces estimated systematic risk. Sector-specific, company-specific, basis, model, and estimation risks remain.

Should Firms Hedge?

Potential benefits

- More predictable cash flows
- Lower financial-distress risk
- Better investment planning
- Focus on core business risks

Potential costs

- Transaction and monitoring costs
- Basis and model risk
- Reduced favourable outcomes
- Contract-size and maturity mismatch

Judge the hedge using the **combined exposure and derivative position**, not the derivative in isolation.

Trading Controls and Rogue-Trader Risk

- A hedge can become a speculative position if its size or purpose changes.
- Trading mandates should define permitted instruments, position limits, and hedge objectives.
- Independent valuation, reporting, and reconciliation help detect unauthorised risk-taking.
- Managers should compare the derivative position with the exposure it is intended to hedge.

Final control question

What exposure is this position hedging, and does its size match that exposure?

Lecture Summary

- Hedgers reduce risk; speculators deliberately take risk; arbitrageurs exploit price inconsistencies.
- A long futures hedge protects a future purchase; a short hedge protects a future sale.
- Basis risk arises because the closing basis is uncertain.
- The minimum-variance hedge ratio is $h^* = \rho\sigma_S/\sigma_F$.
- Index futures can change a portfolio's systematic market exposure.

Core principle

A hedge should be evaluated by how it changes the risk of the complete position, not by whether the derivative makes a profit by itself.